



Brightwell Design Studio

IT Improvement Proposal

A practical, cost-conscious plan to improve speed, reliability, and security



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Current environment

12 employees and 20 end-user devices
Creative audio and video work
Peer-to-peer network with one router
Low-cost cloud storage for shared files

Main complaints

Slow file transfers
Weak Wi-Fi in parts of the building
Long waits during video rendering
Recent breach through unsecured Wi-Fi

Proposal goal

Remove daily bottlenecks
Protect client and company data
Create a setup that can grow with the business



- Creative work depends on fast storage, stable networking, and secure systems.
- Slow systems increase turnaround time, frustrate employees, and delay client delivery.
- Weak security puts client files, company files, and business reputation at risk.
- A stronger IT foundation supports daily work now and future growth later.

What success should look like

Project files open quickly

Editors work with fewer interruptions

Wi-Fi is stable across the office

Unauthorized users stay out



Current standard device

Windows 10 Home Edition
Intel Core i5, 8th generation
8 GB DDR4 memory
500 GB HDD
Integrated Intel UHD graphics

What this means

Adequate for email, web, and light office work
Not ideal for modern creative workloads
Hard drives slow startup, file access, and caching
Integrated graphics limit rendering performance

Business concern

Windows 10 support ended on October 14, 2025
Home edition lacks several business management and security controls
The platform is now behind both on support and on business readiness



What Is Driving the Complaints?

Root causes

Slow file transfers

Project files are moving through one broadband link.

Old HDDs slow reading and writing.

Some users work over Wi-Fi instead of wired Ethernet.

Low-cost cloud storage is not built for heavy local editing workflows.

Weak Wi-Fi coverage

One router is covering the entire building.

Signal strength drops with distance and walls.

No separate wireless access points are extending coverage.

Long rendering waits

8 GB RAM is low for creative apps.

Integrated graphics limits acceleration.

Older i5 processors take longer on heavy exports.

Hard drives increase scratch-disk delays.

Security breach

The Wi-Fi network was unsecured.

An outside user connected from beyond the building.

This exposes the whole office to unauthorized access.



Today

One 802.11ac Wi-Fi router.
Five Ethernet ports.
Shared broadband connection.
Some devices wired, some wireless.



Network effect

Single point of failure.
Inconsistent speeds by location.
No room for clean segmentation.
Limited path for future growth.

Bottom line

The current network can keep a small office online.
It is not strong enough for dependable creative production and secure business use.



For content creators

Windows 11 Pro.

Modern Intel Core i7 or AMD Ryzen 7 class CPU.

16 GB RAM minimum; 32 GB for lead editors.

1 TB NVMe SSD.

Dedicated graphics card.

Why this helps

Much faster boot and file access.

Better multitasking in creative apps.

Shorter export and render times.

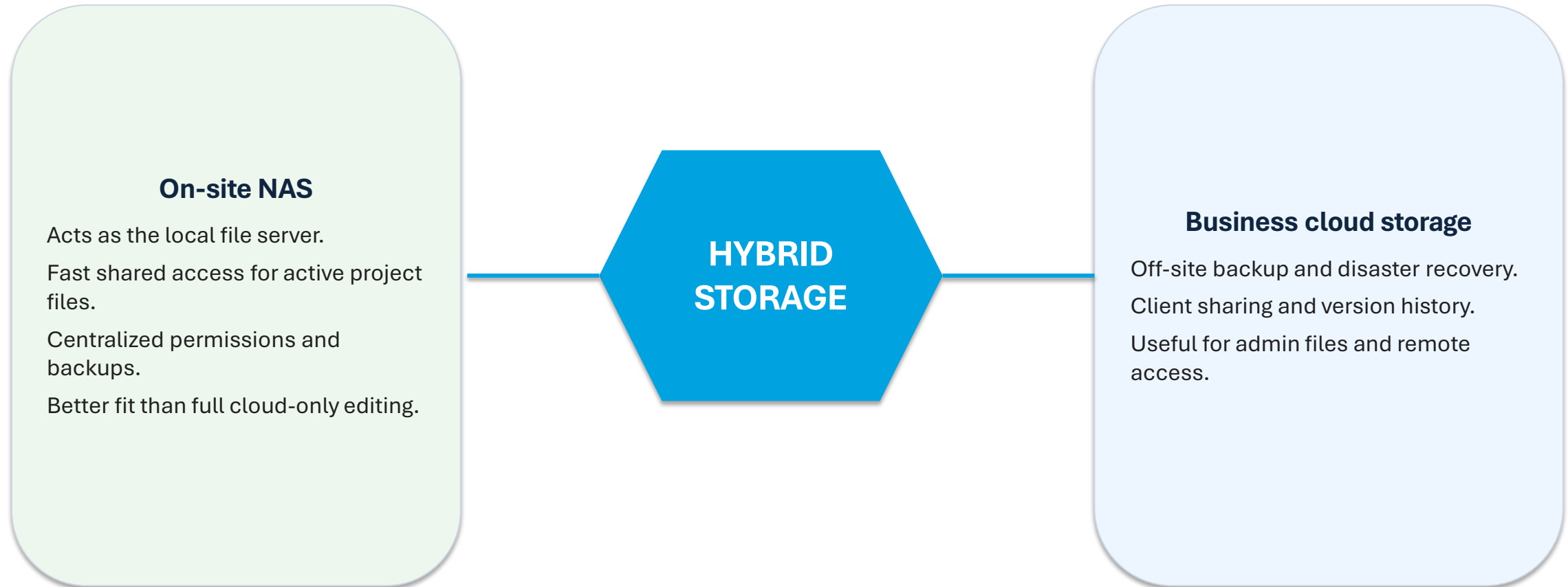
Stronger business security and management controls.

Practical rollout

Upgrade creator devices first.

Keep older PCs for light admin use if needed.

Replace by priority instead of all at once.



Recommendation: Brightwell does not need a full rack server right now. It does need a proper shared storage platform.



Network design

Replace the single consumer router with a business firewall/router.

Add a managed gigabit switch.

Install 2 to 3 Wi-Fi 6 access points for full coverage.

Use wired Ethernet for creator workstations when possible.

Security controls

Use WPA3 Wi-Fi security.

Create separate staff and guest networks.

Disable WPS and change admin defaults.

Keep router and access point firmware updated.

Turn on MFA for cloud accounts.

Use endpoint protection and BitLocker.



Phase 1 — Immediate

Secure Wi-Fi and redesign the network
Separate guest access.
Update firmware and change defaults
Turn on MFA.

Phase 2 — Next

Deploy NAS plus business cloud backup
Move active project files to local shared storage.
Create simple backup and recovery routine.

Phase 3 — Planned refresh

Replace creator systems by priority.
Start with the heaviest video users.
Reuse older machines for light office tasks.



- Faster project access, smoother edits, and shorter render times.
- More dependable Wi-Fi and fewer productivity interruptions.
- Stronger protection for client files and company data.
- Better backup, recovery, and continuity if something goes wrong.
- An IT foundation that supports future growth instead of slowing it down.

Efficiency

Security

Reliability

Scalability



1. Secure the network first

Business firewall/router.
Wi-Fi 6 access points.
WPA3 and guest isolation.

2. Move to hybrid storage

Local NAS for active work.
Cloud for backup and sharing.

3. Refresh creator systems

Windows 11 Pro.
SSD, more RAM, dedicated GPU.

Bottom line: invest once in the right areas so the team can work faster, safer, and with less friction every day.



Thank You

Questions or follow-up
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Microsoft Support. Windows 10 support has ended on October 14, 2025. <https://support.microsoft.com/en-us/windows/windows-10-support-has-ended-on-october-14-2025-2ca8b313-1946-43d3-b55c-2b95b107f281>

Microsoft. Compare Windows 11 for Business editions. <https://www.microsoft.com/en-us/windows/business/compare-windows-11>

CISA. Securing Enterprise Wireless Networks. <https://www.cisa.gov/news-events/news/securing-enterprise-wireless-networks>

Wi-Fi Alliance. Security and WPA3 overview. <https://www.wi-fi.org/security>

Backblaze. NAS vs. Cloud Storage. <https://www.backblaze.com/blog/nas-vs-cloud-storage-which-solution-fits-your-business-needs/>